Large Language Models for Bayesian Network Parameterization in Software Engineering: Direct Elicitation versus Ranked Nodes

Anonymous Author(s)

Abstract

Software engineering (SE) decision makers need models they can understand and use when data are scarce. Bayesian networks (BNs) make probabilistic relationships explicit and incorporate expert knowledge, but eliciting their conditional probability tables (CPTs) is costly. The Ranked Nodes Method (RNM) reduces the required input by generating ordinal CPTs from partial judgments. Large language models (LLMs) offer another source of judgments, raising a question: should they fill complete tables or supply fewer inputs to RNM? We compared these approaches on 61 ordinal conditional nodes from seven SE BNs. Structured prompting and qualitative retrieval-augmented generation (RAG) formed additional factors in eight conditions, each repeated five times using GPT-6 Astra at medium effort. On the unqueried rows of 25 nodes, RNM reduced mean total variation distance by 13.3% for a Scrum process-assessment BN but increased it by 13.2% for two Chrome code-review BNs. These contrasts passed Holm-adjusted paired t-tests, but not exact sign tests. Structured prompting and qualitative RAG provided no consistent benefit. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, but only 0.61% fewer total tokens. Its primary accuracy benefit occurred for a reference itself constructed with RNM, supporting a compatibility explanation rather than a general advantage over direct elicitation. The findings favor checking reconstruction assumptions and numerical agreement before adopting RNM to reduce LLM elicitation effort.

CCS Concepts

• **Computing methodologies** → **Bayesian networks**; • **Software and its engineering** → *Software creation and management*.

Keywords

Bayesian networks, large language models, probability elicitation, ranked nodes, software engineering

1 Introduction

Supporting software engineering (SE) decisions faces two main challenges: explaining the basis for a recommendation and obtaining enough relevant data to build a model experts can rely on. Examples include release readiness, quality assessment, and process improvement, where decision makers must judge whether a recommendation's assumptions apply. A review of 108 studies documents the demand for explainability across 23 SE tasks [3]. Training data for machine learning and deep learning may also be scarce, particularly for managerial SE decisions [23]. Explicit assumptions and domain knowledge are therefore valuable.

Bayesian networks (BNs) offer such an approach. A BN makes relationships and uncertainty explicit in a graph and probabilities, incorporating domain knowledge when data are limited [10, 11]. BNs have supported many SE problems, including early defect prediction [10], effort estimation [17], and diagnosis of development-process problems [21]. A systematic mapping of 106 studies identifies software management and quality as prominent application areas [24]. Building a credible BN nevertheless requires substantial knowledge-engineering effort.

Constructing conditional probability tables (CPTs) is a major part of that effort. Each row describes a child's probabilities for one combination of its parents' states; adding parents rapidly increases the number of judgments required [11]. Semi-automatic methods reduce this burden by reconstructing a table from fewer expert judgments or parameters. These include the Ranked Nodes Method (RNM) [11], the Weighted Sum Algorithm (WSA) [20], Analytic Hierarchy Process (AHP)-based pairwise comparisons [5], and interpolation methods [1, 2]. Their assumptions can, however, limit fidelity to the intended probabilities [2].

Large language models (LLMs) are now being explored for CPT construction. Nafar et al. elicited numerical BN parameters directly [18]; Gottal and Matthes reconstructed CPTs from statistical summaries extracted from scientific literature [12]; and Cypko et al. used retrieval-augmented generation (RAG) to support expert BN modeling [6]. Should an LLM specify every row, or supply fewer judgments for a semi-automatic method to complete? Any benefit depends on both the judgments and the reconstruction.

We investigated this question using RNM, which applies to variables with ordered states such as Low, Medium, and High [11, 15]. Instead of requesting the entire CPT, we asked the LLM for distributions at selected parent-state combinations, called *anchors*, and for a rule describing how the parents combine. A numerical fitting procedure then adjusted that rule's weights and uncertainty to match the elicited distributions and generated the complete CPT. We compared this approach with direct complete-table elicitation on 61 ordinal conditional nodes from seven SE BNs.

We also examined support for elicitation. Structured prompts made conditioning explicit and asked for probability and consistency checks; qualitative RAG supplied short passages explaining relevant domain relationships. Crossing the two elicitation methods, two prompt designs, and presence or absence of RAG produced eight conditions. We repeated each five times using GPT-6 Astra at medium effort. This study addresses three research questions:

- **RQ1—Reference agreement:** Does RNM reduce total variation distance relative to direct elicitation on the same non-anchor configurations?
- **RQ2—Elicitation support:** How do structured prompting and qualitative RAG affect each method's agreement across complete tables?
- **RQ3—Accuracy and effort:** How does RNM trade reference agreement against requested judgments and observed computational effort?

SAC '27, Gwangju, South Korea
2027.

RQ1 tests whether RNM can complete rows whose distributions were not elicited from the LLM. We required at least two parents and one such row, yielding 25 targets. Their anchors were used to fit RNM, so the primary comparison isolates completion beyond those inputs; anchor agreement and complete-table results are reported separately. Fourteen targets come from a published Scrum process-assessment BN relating teamwork, backlog, and sprint factors [19]. The other 11 come from two Chrome BNs connecting code-review practices, contributor activity, and post-release defects [14]. All 61 targets contribute to complete-table (RQ2) and effort analyses (RQ3).

RNM improved agreement for the RNM-derived Scrum reference but worsened it for the stored Chrome CPTs. Alongside inconsistent prompt/RAG benefits and small total-token savings, this contrast does not support routinely preferring RNM over direct elicitation. Our contribution distinguishes the LLM's elicited judgments from the tables generated by fitting and evaluates their accuracy–effort tradeoff.

Section 2 positions the study in related work, Section 3 describes the experiment, and Section 4 reports the results. Section 5 discusses the RQs and implications for SE, Section 6 examines validity threats, and Section 7 concludes the paper.

2 Related Work

Table 1 positions our comparison by what the LLM supplies and how its output is evaluated. Numerical CPT agreement, modeling support, and downstream predictive performance address different questions.

2.1 Numerical elicitation versus modeling support

Nafar et al.'s broad benchmark is the closest numerical comparison [18]. Requesting one child-state probability at a time lowered median BN KL divergence for seven of eight models, at the cost of more queries. Their FullDist baseline requests one conditional row per query; ours requests the whole CPT. Their granularity result does not determine whether omitting entire configurations is beneficial. Gottal and Matthes instead reconstruct from numerical summaries [12], which impose constraints absent from our qualitative RAG packets. Rahul and Chowdhury's persona-panel preprint reports effort and internal consistency, including refitting self-generated data [22]; this cannot establish independent reference agreement.

BIRD combines neural and symbolic reasoning [7] by combining LLM judgments, constrained fitting, and BN inference [9]. Its downstream evaluation differs from exact CPT recovery. Clinical RAG assists modelers [6], while AutoElicit evaluates prediction after updating elicited parameter priors [4]. Our endpoint is numerical agreement, not explanation quality or practitioner benefit.

2.2 Partial elicitation and our contribution

RNM constructs ordinal CPTs from weighted parent functions and truncated-normal distributions [11, 15]; InterBeta offers another partial-elicitation approach [1]. Blomaard et al. compare RNM, functional interpolation, and InterBeta using known CPTs to optimize parameters [2]. A recent preprint compares human-elicited WSA and RNM in software decisions [20]. Our experimental fits instead use LLM anchors alone; reference-informed fits remain separate diagnostics.

The unresolved comparison is whether sparse LLM judgments plus reconstruction retain agreement while reducing effort. We compare the pipelines on identical unqueried rows and separately compare raw and fitted judgments on identical anchors. This distinguishes elicitation from fitting without assuming that ordinal eligibility guarantees RNM compatibility or that fewer distributions imply proportional token savings.

3 Method

We compared direct CPT elicitation with LLM-assisted RNM reconstruction in a repeated $2 \times 2 \times 2$ design: method (direct/RNM), prompt (baseline/structured), and evidence (no RAG/qualitative RAG). We hypothesized lower non-anchor error with RNM (RQ1), improved agreement from elicitation support (RQ2), and fewer requested distributions and output tokens with RNM (RQ3). Graphs, definitions, state mappings, prompts, and numerical procedures were fixed before collection.

3.1 Sample and references

We selected 61 conditional targets from seven published SE BNs with complete numerical references and interpretable, ordinal child and parent states (Table 2). Binary variables qualified only when their meanings supported an order; nominal families and Boolean gate chains were excluded. This availability-based sample covers Scrum practices [19], Chrome code review and defects [14], ERP data quality for advanced planning and scheduling (APS) [13], software maintenance logs [8], and software strategic indicators (SESSI) [16]. The two Chrome versions share one publication family, as do the two log BNs.

The Scrum reference implements a published framework relating team and process factors to sprint and backlog quality [19]. Its archived XML has 25 conditional nodes, 17 roots, and three states, whereas the publication describes 44 nodes and five states. We reconstructed 249 conditional rows from the XML's functions, weights, variances, and expressions. Scrum is therefore an RNM-derived reference; the other BNs provide exported CPTs, with SESSI using the original elicited table in Appendix A3. We checked dimensions, normalization, and indexing. The largest retained CPT has 27 rows. The supplement specifies reconstruction details.

RQ1 tests CPT rows generated by RNM without eliciting their distributions directly from the LLM. Requiring at least two parents and one such row yielded 14 Scrum and 11 Chrome targets. Direct elicitation supplied LLM distributions for those same rows. The remaining 36 targets had only one parent or no unqueried rows; they contributed to complete-table and effort analyses. References describe particular modeling contexts, not universal company probabilities.

Anchors were excluded only from the primary RQ1 endpoint to isolate completion beyond the elicited inputs. Including them would mix that task with fitting distributions already supplied to RNM. The fitted inputs were LLM judgments, not reference probabilities, so anchor-reference agreement remains informative. We therefore report raw and fitted predictions against the reference

Table 1: LLM-assisted BN construction compared by inputs, evaluation, and scope. Entries distinguish numerical CPT agreement from modeling support and downstream performance; reported errors are not directly comparable. KL denotes Kullback–Leibler divergence.

Study	Construction and supplied evidence	Evaluation material and endpoint	Scope relative to our comparison
Nafar et al. [18] TMLR, 2026	LLM internal knowledge; one conditional row or one child-state probability per query. A separate stage combines estimates with observations.	80 public BNs, eight LLMs: BN/CPT KL. Separate classification evaluation on nine datasets with three LLMs.	Changes granularity within a row; our RNM approach requests selected rows and reconstructs the rest.
Gottal and Matthes [12] Findings ACL, 2026	Reconstruction algorithms and direct LLM CPT output from numerical statistical summaries; separate extraction from publications.	Synthetic BNs and publication/raw-data cases; CPT distances, inference and decision utility, and runtime.	Quantitative summaries constrain construction; our RAG packets supply qualitative context.
Rahul and Chowdhury [22] Preprint, 2026	Persona-conditioned LLM panels provide distributions for parent configurations; trimmed-mean aggregation.	One healthcare-consultation BN; plausibility and internal consistency checks; calls, token estimates, and runtime.	Demonstrates construction feasibility and effort; lacks an independent numerical CPT benchmark.
Feng et al. (BIRD) [9] ICLR, 2025	LLM-generated factors and coarse likelihood judgments feed constrained parameter fitting and BN inference.	Human judgments of relative support and downstream task accuracy.	Establishes a precedent for LLM judgments plus fitting; does not test numerical reference-CPT recovery.
Cypko et al. [6] IJCARS, 2026	Clinical RAG supports expert BN modeling using guidelines and textbooks.	Synthetic retrieval evaluation; modeling, usability, and workload study with four clinicians.	Tests modeling support rather than complete numerical CPT agreement.
This study	Complete CPT elicitation versus LLM anchors plus RNM fitting; baseline/structured prompts, with/without qualitative RAG.	61 ordinal conditional nodes from seven SE BNs; 25 primary targets; reference distances, five replications, tokens, and fitting time.	Compares both pipelines on identical non-anchor rows; separately compares raw and fitted judgments on identical anchor rows.

Table 2: Retained targets and requested rows per prompt/evidence combination and replication. Primary targets have at least two parents and at least one non-anchor row.

Network	Targets	Primary	CPT rows	Anchors
Scrum	25	14	249	123
Chrome 40	15	6	78	57
Chrome 42	12	5	63	46
APS	3	0	8	8
SESSI	1	0	3	3
Software logs 3	4	0	16	12
Software logs 4	1	0	4	3
Total	61	25	421	252

on identical anchors, alongside complete-table TV including anchors. These complementary analyses assess judgment quality, the changes introduced by fitting, and the final CPTs.

3.2 What the LLM received and supplied

All conditions shared an expert-role description, setting, variable definitions, state labels and order, and assumed parent influence directions: whether higher parent states should tend to increase or decrease the child. Each distribution conditioned on *all* listed parent states and averaged over unspecified circumstances. The persona supplied a professional perspective without claiming company observations.

Baseline prompts requested normalized distributions in JSON. Structured prompts additionally made conditioning explicit, framed probabilities as expected counts among 100 hypothetical comparable cases, distinguished evidence from assumptions, and requested consistency checks. Fractional counts were allowed; these cases aided judgment rather than supplying observations. A brief evidence note was requested, without private reasoning traces.

Direct elicitation requested every CPT row in one call. RNM requested predetermined anchor distributions and a combination rule: weighted mean, weighted minimum, weighted maximum, or minimum–maximum mixture. These express different degrees of compensation between stronger and weaker parents. A numerical fitter then estimated weights and variance from the LLM's answers. With one parent, its direction-aligned value supplied the distribution mean directly. Both methods received the same influence assumptions; only RNM enforced them mathematically. Of 92 directions, nine followed from definitions, 43 had qualitative support, and 40 were labeled uncertain hypotheses.

RAG supplied fixed cards paraphrasing relevant domain explanations. One complete card for Scrum Cohesion stated:

> **Shared goals.** Cohesive teamwork includes commitment to shared tasks and team goals. Agile collaboration involves joint responsibility, rather than treating every task as an isolated individual obligation.

This adds context, not frequencies for Low, Medium, or High cohesion. Retrieval matched cards to model context and target/parent tags, selecting up to three; packets contained 24–103 words (median 72). Each node's packet was reused across paired conditions and replications. Cards excluded reference probabilities, numerical parameters, and identifying artifact links. Both arms already received definitions, so RAG tested additional qualitative context, not live retrieval or company measurements. Input budgets were not equalized.

3.3 From anchors to complete tables

RNM mapped ordered states to equal intervals in $[0, 1]$, reversed negative parent influences, and averaged truncated-normal probabilities over five integration points per parent interval [11, 15]. Anchors comprised all-low, all-high, all-middle, one-high/rest-low, and one-low/rest-high configurations, removing duplicates without consulting the reference. Two ternary parents, for example, yield nine rows and five anchors: $(0, 0)$, $(2, 2)$, $(1, 1)$, $(2, 0)$, and $(0, 2)$. RNM generated the other four rows; RQ1 compared them with the LLM distributions elicited directly for those same rows.

For the LLM-selected rule, deterministic multistart search minimized mean anchor TV between elicited and fitted distributions, without reference access. Valid poor fits were retained. The supplement provides equations, parameter bounds, integration, and search settings. Agreement does not establish recovery of unique parameters.

3.4 Execution and reference isolation

Five replications were collected on 15–16 October 2026 using Codex CLI 0.162.1, requesting `gpt-6-astra` at `medium` effort. Each target–condition–replication used a fresh session; requests were shuffled within replications. Temperature and generation seed were unspecified. Experimental sessions could not access references, full papers, evaluator code, or authoring conversations; tools, browsing, connectors, and memory were disabled.

Outputs required correct rows/states, finite probabilities in $[0, 1]$, and sums within 10^{-6} of one. One fresh-session format repair was allowed without reference feedback. All 2,440 trials produced valid outputs after four repairs (two per method); accounting includes all 2,444 attempts, with no fallback distributions.

3.5 Outcomes and analysis

For reference row p and prediction q, lower total variation means closer agreement:

$$\mathrm{TV}(p, q) = \tfrac{1}{2} \sum_{s} |p_s - q_s| \in [0, 1]. \tag{1}$$

RQ1 scored identical non-anchor rows, averaging rows within targets, targets within BNs, and BNs within publication families, then equally averaging the four prompt/evidence combinations in each replication. Scrum and Chrome were reported separately, with equal weight for Chrome versions. RQ2 contrasted structured minus baseline and RAG minus no RAG on complete tables, averaging the other factor; its external SE mean weighted four publication families equally. Secondary distances and full condition means appear in the supplement.

RQ3 counted requested vectors before repairs and client-reported tokens over all attempts. Total tokens equal input plus output; cached-input and reasoning-output counters are subsets, not additional tokens. Paired contrasts used replication totals for all 61 and the 25 primary targets. Latency and offline fitting time were descriptive; subscription tokens were not converted into monetary savings.

The five paired replication blocks were the inferential units, not correlated nodes or rows. Two-sided paired t-tests ($df = 4$) and pointwise 95% intervals assumed approximately normal, independent, stationary block differences. Five blocks provide little evidence for these assumptions. Two-sided exact sign tests assessed direction/median sensitivity under independence; five concordant nonzero differences still yield $p = 0.0625$. Holm adjustments were separate for two primary, eight prompt/RAG, four token, and two same-anchor fitting contrasts. Intervals were not simultaneous; other comparisons were descriptive. No prespecified practical-effect or equivalence margin justified claims of usefulness or equivalence.

3.6 Offline diagnostics

Post-hoc analyses used saved responses to compare raw and fitted anchor agreement, uniform predictions, alternative aggregation-family choices, direct judgments on paired RNM anchors, stronger search, and wider bounds. These fits retained reference isolation. Separate reference-informed fits assessed best-found approximation under altered bounds and directions, not held-out accuracy or certified error floors. Chrome-version and leave-one-node-out analyses tested concentration of effects. The supplement reports procedures and results; none replaced the original predictions.

4 Results

RNM requests fewer judgments, but its effect on agreement depends on the reference. We examine reconstruction, elicitation support, and effort in turn. Lower TV indicates closer agreement; negative RNM-minus-direct differences favor RNM.

4.1 RQ1: When does reconstruction improve agreement?

The same method helps Scrum and hurts Chrome. On the 14 primary Scrum targets, mean non-anchor TV decreases from 0.3184 with direct elicitation to 0.2761 with RNM: a reduction of 0.0423, or 13.3%. RNM is closer on 12 of the 14 targets and in each of the four matched prompt/evidence conditions. On the 11 Chrome targets, the result reverses: TV increases from 0.2223 to 0.2517, a rise of 0.0294, or 13.2%. RNM is closer on only three targets. Thus, the hypothesized benefit holds for the RNM-derived Scrum reference, but not for the independently stored Chrome CPTs.

Figure 1a shows that this contrast is present in every replication, rather than being caused by one unusual run. The paired-t intervals for RNM minus direct are $[-0.0486, -0.0360]$ for Scrum and $[+0.0259, +0.0329]$ for Chrome; both Holm-adjusted p-values are below 0.001. However, the exact sign tests do not reject after adjustment ($p = 0.125$ in each stratum). RNM improves Scrum and worsens Chrome in every replication, while statistical rejection depends on the paired-t assumptions. These tests do not determine whether the differences are useful for a particular SE decision.

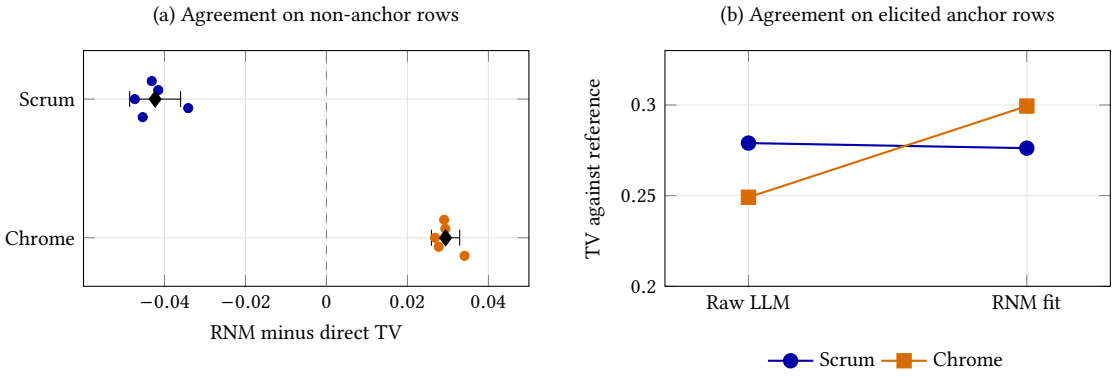


Figure 1: Where the methods diverge. (a) Circles show five replication differences on non-anchor rows, averaging prompt/evidence conditions for 14 Scrum or 11 Chrome targets. Diamonds and whiskers show means and pointwise 95% t intervals; vertical jitter separates runs. Negative values favor RNM. (b) TV before and after fitting the same anchor rows. The panels use different row sets and do not decompose total error.

The anchor rows locate a problem in reconstruction. Anchors are excluded from the primary completion score but remain important to the final CPT. Comparing raw and fitted distributions on the *same anchor rows* identifies what fitting changes (Figure 1b). For Scrum, raw-anchor TV is 0.2790 and fitted-anchor TV is 0.2762: fitting changes agreement little. For Chrome, it rises from 0.2491 to 0.2994. Direct judgments on these Chrome anchors have TV 0.2507, close to the raw RNM value. On the elicited Chrome configurations, fitting introduces additional disagreement beyond that present in the raw judgments.

The contrast is particularly pronounced in Chrome 42. Its primary TV rises from 0.2377 to 0.2933, whereas Chrome 40 changes from 0.2069 to 0.2102. Removing any one target preserves the Scrum benefit or Chrome disadvantage in all five replications. The next examples show what these changes can look like in a CPT; they illustrate mechanisms rather than typical cases.

Two configurations make the tradeoff concrete. Consider a Scrum team with low self-organization but medium collaboration. The model's Cohesion node asks whether the team works synchronously toward shared goals. For this parent configuration, the reference assigns 76.6% probability to Low cohesion. Direct elicitation assigns 58.0%, while RNM reconstructs 67.6%, moving closer to the reference (Figure 2a). Mean row TV falls from 0.1862 to 0.0901. This is a configuration that the RNM prompt did not ask the LLM to fill: the fitted relationship supplies it from the anchors.

Now consider a Chrome component with three or four contributors and no minor contributors. Contributors are classified as major or minor according to their share of changes. With no minor contributors, all must be major, so the Major Contributors node belongs entirely in the three-to-four category. Both the reference and all direct responses assign probability one to that category. The RNM reconstruction assigns 84.4%, with the remaining 15.6% placed in neighboring categories (Figure 2b). Here, smoothing a table through the fitted family weakens a count constraint that direct elicitation preserves. This target has the largest Chrome disadvantage, but removing it still leaves Chrome's aggregate difference positive at +0.0168.

Alternative fits retain the contrast. Fitting the direct method's judgments on paired RNM anchors still improves Scrum and worsens Chrome. Choosing families by anchor fit narrows Chrome's disadvantage; stronger search and wider bounds do not remove it. Reference-informed fits also approximate Scrum more closely, even when bounds and directions vary. These checks support a compatibility limitation in the tested pipeline rather than a simple shortage of search effort, without isolating every component or establishing an error floor. Detailed values and direction-reversal diagnostics are in the supplement.

Across all 61 targets, complete-table TV changes from 0.2605 to 0.2334 for Scrum and from 0.1770 to 0.2153 for the four-family external SE mean. Chrome, APS, and software logs worsen; SESSI improves slightly. The supplement provides full condition means, family results, baselines, and sensitivity analyses. **RQ1 is supported for Scrum and opposed for Chrome.** Ordinal eligibility alone does not guarantee better agreement.

4.2 RQ2: Do richer prompts or qualitative evidence help?

The method contrast remains across the prompt/evidence combinations. Table 3 isolates the incremental effect of structured prompting and RAG on complete-table TV. For direct Scrum elicitation, structured prompting lowers TV by 0.0069, while RAG raises it by 0.0043; these are the only two comparisons surviving the eight-test Holm correction. Neither RNM Scrum contrast nor any external SE contrast survives that correction, and none rejects under the exact sign test.

The Cohesion packet, for example, describes shared goals and joint responsibility without saying how often cohesion is Low given Low self-organization. RAG therefore adds qualitative guidance to shared definitions, without numerical constraints. **RQ2 receives**

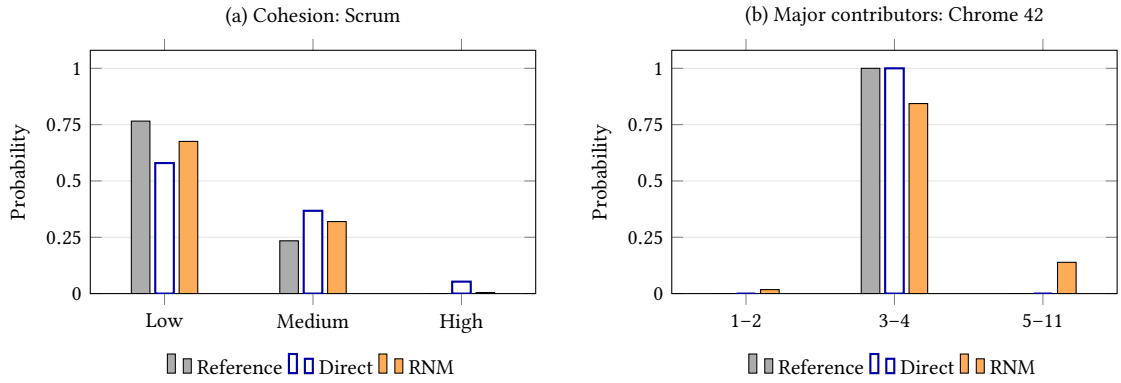


Figure 2: Two CPT rows unqueried by RNM. (a) Cohesion given Low Self-Organizing and Medium Collaboration. (b) Major contributors given three or four total contributors and no minor contributors. Bars average 20 predictions per method. Illustrations use the targets with the largest improvement and deterioration and their first stored non-anchor rows; they are deliberately contrasting, not representative.

Table 3: RQ2: complete-table TV change from adding structured prompting or RAG, averaging the other factor equally. Negative values favor the addition. Asterisks mark Holm-adjusted paired-t $p < 0.05$ across eight contrasts; no exact sign test rejects. Full intervals and p-values are in the supplement.

Reference	Method	Prompt Δ	RAG Δ
Scrum	Direct	−0.00692*	+0.00430*
Scrum	RNM	−0.00313	+0.00132
External SE	Direct	−0.00462	+0.00163
External SE	RNM	−0.00060	+0.00085

no consistent support for the tested prompt bundle and qualitative packets. This result does not establish that company measurements or quantitative retrieval would be ineffective.

4.3 RQ3: Do fewer judgments translate into fewer tokens?

RNM requests 5,040 distributions across the experiment, compared with 8,420 for direct elicitation: 40.1% fewer. Yet both methods still make one initial request per target and condition. Fewer table entries therefore shorten what must be returned, without eliminating the shared instructions, definitions, or evidence sent with each request.

This distinction explains why the savings shrink as accounting broadens. Output tokens decrease by 12.12%, but input-plus-output tokens decrease by only 0.61% (Table 4). Inputs dominate the total: approximately 14.36 million for direct elicitation and 14.31 million for RNM. The 25 primary targets, where multi-parent compression is possible, show larger savings: 28.28% in output tokens and 2.94% in total tokens. Even there, the reduction in requested probabilities is not a proportional reduction in input-plus-output tokens.

Output savings survive the paired-t adjustment for both workloads, and total-token savings survive it for the primary targets only. Summed request latency is 3.82% lower for RNM; offline reconstruction adds 182.3 seconds of recorded fit time. These are computational descriptions, not controlled wall-clock or monetary savings. **RQ3 is supported for reduced elicitation and output size, but offers limited evidence of overall token savings.** The observed token savings accompany improved Scrum agreement but worse Chrome agreement: fewer requested judgments do not ensure a better CPT.

5 Discussion

The results connect two questions that are often considered separately: whether an LLM can supply useful probability judgments and whether a compact construction method preserves them. The contrast between Scrum and Chrome shows why the complete construction process matters. We discuss that contrast against prior work, then draw implications for developing and applying SE BNs.

5.1 Research questions in relation to the literature

RQ1: Reference agreement. For RQ1, RNM's improvement on Scrum is consistent with its ability to represent a reference already constructed using ranked nodes [11]; the improvement also persisted when fitting used the direct method's distributions at the same anchor configurations. Chrome's raw anchor judgments are similarly accurate across methods, but fitting worsens agreement. The disadvantage concentrates in Chrome 42; no single target determines its direction. Choosing the combination rule by numerical fit to the LLM judgments narrowed this disadvantage, but a more thorough search and wider parameter bounds did not remove it. The primary accuracy benefit is therefore confined to the RNM-derived reference, consistent with compatibility between its construction and the fitted family [15]. Reference origin was not experimentally varied, so it cannot alone be established as the cause. Reference-informed direction changes reduce approximation error, but cannot separate remaining representation and optimization limitations. Chrome's documented interdependencies [14] provide context, not

Table 4: RQ3: observed tokens over all five replications, including format repairs. Workloads combine reference strata for accounting only. Holm-adjusted paired-t tests compare five method-paired replication totals across four contrasts; no exact sign test rejects. Complete intervals are in the supplement.

Targets	Counter	Direct	RNM	Reduction	Holm t p
All 61	Output	341,153	299,798	12.12%	0.000110
All 61	Input + output	14,697,090	14,607,517	0.61%	0.080921
Primary 25	Output	213,972	153,463	28.28%	0.000019
Primary 25	Input + output	6,384,519	6,196,992	2.94%	0.016292

proof of the mechanism. Nor does CPT agreement establish defect-prediction accuracy.

RQ2: Elicitation support. Structured prompting benefits direct Scrum elicitation but not the other adjusted comparisons. Nafar et al. likewise distinguish query-granularity gains from prompt modifications [18]; they changed how individual row probabilities were requested, whereas we reduced the number of rows requested. Their results therefore cannot establish the benefit of RNM reconstruction. Qualitative packets give no consistent benefit beyond shared definitions. Clinical retrieval-usability findings [6] and reconstruction from statistical summaries [12] address different evidence and outcomes. Our results do not establish that informative quantitative retrieval is ineffective.

RQ3: Accuracy and effort. Fewer distributions yield lower output use, but small total-token savings. Extending the human burden–fidelity perspective [1], computational effort includes inputs, reasoning, repairs, and fitting. Compression improves reference agreement on Scrum while worsening it on Chrome, preventing a universal efficiency–accuracy recommendation.

5.2 Implications for SE researchers and practitioners

For researchers, the contrast motivates independently constructed references, validated influence directions, and multiple model configurations. Comparing raw and fitted distributions on the same anchors shows what fitting changes; evaluating unqueried rows tests whether those changes extend usefully to the rest of the table. Hybrid inference such as BIRD [9] also motivates testing structured components rather than inferring fidelity from inspectability. Repeated runs do not establish generalization to other companies.

For practitioners, these results do not support replacing direct elicitation with RNM by default. Direct elicitation retained better agreement on the primary external CPT comparison; RNM remains a candidate when its assumptions fit the intended model. Validate meanings, influence relationships, and additional configurations with company measurements or experts. A small fitting error only means that the reconstructed table matches the elicited judgments; it does not establish that those judgments describe the organization. Tailoring reusable models remains plausible but requires held-out organizational validation before decision use.

6 Threats to Validity

Construct validity. TV measures contextual CPT agreement, not observed calibration, expert replacement, explanation quality, or decision utility. Equal-row scoring includes potentially rare configurations. Scrum's reconstruction lacks an independently certified AgenaRisk export, and its RNM origin favors the tested representation. Separate strata and anchor diagnostics expose this distinction. SESSI represents an original elicitation stage. Adapted variable meanings and uncertain parent directions may affect agreement despite correct indexing.

Internal validity. Fresh sessions, shared definitions, reference isolation, and blinded fitting limit direct access leakage. Public benchmark memorization remains possible. The logs identify the requested model configuration, but cannot independently verify the exact serving-model version. Prompt and fitting development used the Scrum setting, which may favor that reference. Offline variants cannot exhaust all explanations. Reference-informed selection is optimistic in-sample diagnosis; bounded search may miss better fits. Schema validation cannot establish semantic or evidence fidelity, and exhaustive human review was not performed. The elicitation instructions and evidence cards were tested as bundles, so the contribution of each individual instruction or passage is unknown.

Conclusion validity. Five blocks give little evidence about the distribution or stationarity of differences. The paired-t and exact-sign results must be considered together; neither treats shared nodes or rows as independent replications. The SE subset and statistical comparisons were selected after inspecting the broader results; inference is therefore exploratory. Separate Holm families do not provide study-wide confirmatory error control. No prespecified acceptable-error margin supports claims of equivalence or practical sufficiency. Repeated execution estimates here are conditional on these fixed references and settings.

External validity. The primary comparison contains one external publication family (Chrome) and one Scrum model. The 61-target sample spans more settings, but many targets do not test multi-parent compression, and tables have at most 27 rows. Only one requested model and reasoning effort were evaluated. Organizational distributions, larger CPTs, other model families, and evidence containing local numerical measurements may yield different trade-offs.

7 Conclusion

Across 61 ordinal conditional nodes from seven SE BNs, we compared complete LLM elicitation with selected judgments followed by RNM reconstruction, under eight conditions and five replications. The results do not support routinely preferring RNM over direct elicitation for this accuracy–effort tradeoff.

On the 25 targets with rows generated by RNM without direct LLM judgments, RNM reduced TV by 13.3% for Scrum's RNM-derived reference and increased it by 13.2% for Chrome. Raw Chrome anchors were similarly accurate across methods, but RNM fitting increased disagreement. The contributor-count example illustrates how fitting can lose a constraint preserved by direct elicitation. Alternative fits and single-node omissions retained the Scrum benefit and Chrome disadvantage, supporting a compatibility limitation beyond one exceptional target. The design does not establish reference construction alone as its cause. Structured prompting and qualitative RAG offered no consistent improvement.

RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, yet input-dominated workloads left only 0.61% fewer total tokens. Researchers should therefore evaluate judgment quality, reconstruction, and resource use together. Practitioners can treat LLM-generated CPTs as candidate initializations requiring validation in their organizational context; ordinal labels alone do not establish RNM suitability.

These findings concern reference agreement with one LLM configuration, not validated organizational decisions. Scrum improved and Chrome worsened in all five replications. Both contrasts survived Holm-adjusted paired t-tests, but neither exact sign test rejected. Future work should test domain-validated influences and organizational measurements, then assess prediction and decision quality on held-out projects. This connects numerical fidelity to the practical purpose of SE decision support.

Replication Materials

The replication package contains the elicitation materials, reference CPTs, saved responses, fitted tables, evaluation scripts, token counts, and supplementary analyses. Reference materials are separated from collection inputs. The repository is private for author review.

Use of Generative AI

LLM judgments are experimental data (Section 3). Generative AI assisted drafting and script preparation/checking; numerical claims trace to saved responses and deterministic analyses.

References

[1] Martine J. Barons, Steven Mascaro, and Anca M. Hanea. 2022. Balancing the Elicitation Burden and the Richness of Expert Input When Quantifying Discrete Bayesian Networks. *Risk Analysis* 42, 6 (2022), 1196–1234. doi:10.1111/risa.13772

[2] Bodille P. M. Blomaard, Gabriela F. Nane, and Anca M. Hanea. 2025. Approaches for Reducing Expert Burden in Bayesian Network Parameterization. *Entropy* 27, 6, Article 579 (2025), 27 pages. doi:10.3390/e27060579

[3] Sicong Cao, Xiaobing Sun, Ratnadira Widyasari, David Lo, Xiaoxue Wu, Lili Bo, Jiale Zhang, Bin Li, Wei Liu, Di Wu, and Yixin Chen. 2025. A Systematic Literature Review on Explainability for ML/DL-based Software Engineering. *Comput. Surveys* 58, 4, Article 95 (2025), 34 pages. doi:10.1145/3763230

[4] Alexander Capstick, Rahul Krishnan, and Payam Barnaghi. 2025. AutoElicit: Using Large Language Models for Expert Prior Elicitation in Predictive Modelling. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 6746–6777. https://proceedings.mlr.press/v267/capstick25a.html

[5] Kwai-Sang Chin, Da-Wei Tang, Jian-Bo Yang, Shui Yee Wong, and Hongwei Wang. 2009. Assessing new product development project risk by Bayesian network with a systematic probability generation methodology. *Expert Systems with Applications* 36, 6 (2009), 9879–9890. doi:10.1016/j.eswa.2009.02.019

[6] Mario A. Cypko, Muhammad Agus Salim, Aditya Kumar, Leonard Berliner, Andreas Dietz, Matthaeus Stoehr, and Oliver Amft. 2026. Large language models with retrieval-augmented generation enhance expert modelling of Bayesian network for clinical decision support. *International Journal of Computer Assisted Radiology and Surgery* 21, 2 (2026), 211–222. doi:10.1007/s11548-025-03524-9

[7] Artur d'Avila Garcez and Luis C. Lamb. 2023. Neurosymbolic AI: the 3rd wave. *Artificial Intelligence Review* 56, 11 (2023), 12387–12406. doi:10.1007/s10462-023-10448-w

[8] Santiago del Rey, Silverio Martínez-Fernández, and Antonio Salmerón. 2023. Bayesian Network analysis of software logs for data-driven software maintenance. *IET Software* 17, 3 (2023), 268–286. doi:10.1049/sfw2.12121

[9] Yu Feng, Ben Zhou, Weidong Lin, and Dan Roth. 2025. BIRD: A Trustworthy Bayesian Inference Framework for Large Language Models. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=fAAaT826Vv

[10] Norman Fenton, Martin Neil, William Marsh, Peter Hearty, Łukasz Radliński, and Paul Krause. 2008. On the effectiveness of early life cycle defect prediction with Bayesian Nets. *Empirical Software Engineering* 13 (2008), 499–537. doi:10.1007/s10664-008-9072-x

[11] Norman E. Fenton, Martin Neil, and Jose Galan Caballero. 2007. Using Ranked Nodes to Model Qualitative Judgments in Bayesian Networks. *IEEE Transactions on Knowledge and Data Engineering* 19, 10 (2007), 1420–1432. doi:10.1109/TKDE.2007.1073

[12] Jonas Gottal and Florian Matthes. 2026. Verifiable Parameterization of Bayesian Networks from Scientific Literature: Unlocking Unstructured Empirical Evidence. In *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, 3400–3414. doi:10.18653/v1/2026.findings-acl.167

[13] Jan-Phillip Herrmann, Sven Tackenberg, Elio Padoano, Jörg Hartlief, Jens Rautenstengel, Christine Loeser, and Jörg Böhme. 2022. An ERP Data Quality Assessment Framework for the Implementation of an APS system using Bayesian Networks. *Procedia Computer Science* 200 (2022), 194–204. doi:10.1016/j.procs.2022.01.218

[14] Andrey Krutauz, Tapajit Dey, Peter C. Rigby, and Audris Mockus. 2020. Do code review measures explain the incidence of post-release defects? Case study replications and Bayesian networks. *Empirical Software Engineering* 25 (2020), 3323–3356. doi:10.1007/s10664-020-09837-4

[15] Pekka Laitila and Kai Virtanen. 2020. On Theoretical Principle and Practical Applicability of Ranked Nodes Method for Constructing Conditional Probability Tables of Bayesian Networks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 50, 5 (2020), 1943–1955. doi:10.1109/TSMC.2018.2792058

[16] Martí Manzano, Claudia Ayala, Cristina Gómez, Antonin Abherve, Xavier Franch, and Emilia Mendes. 2021. A Method to Estimate Software Strategic Indicators in Software Development: An Industrial Application. *Information and Software Technology* 129 (2021), 106433. doi:10.1016/j.infsof.2020.106433

[17] Emilia Mendes. 2012. Using knowledge elicitation to improve Web effort estimation: Lessons from six industrial case studies. In *Proceedings of the 34th International Conference on Software Engineering*. IEEE, 1112–1121. doi:10.1109/ICSE.2012.6227108

[18] Aliakbar Nafar, Kristen Brent Venable, Zijun Cui, and Parisa Kordjamshidi. 2026. Extracting Probabilistic Knowledge from Large Language Models for Bayesian Network Parameterization. *Transactions on Machine Learning Research* (May 2026). https://openreview.net/forum?id=Fy3Byg3CVo

[19] Mirko Perkusich, Kyller Gorgonio, Hyggo Almeida, and Angelo Perkusich. 2017. A Framework to Build Bayesian Networks to Assess Scrum-based Software Development Methods. In *Proceedings of the 29th International Conference on Software Engineering and Knowledge Engineering*. 7 pages. doi:10.18293/SEKE2017-139

[20] Mirko Perkusich, João Nunes, Emilia Mendes, Emanuel Dantas, Ademar Sousa, Danyllo Albuquerque, Kyller C. Gorgônio, and Angelo Perkusich. 2026. Semi-Automatic Quantification of Bayesian Networks for Software Decision Support: Comparing WSA and RNM in a Software R&D Organization. *arXiv preprint arXiv:2609.31275* (2026), 37 pages. https://arxiv.org/abs/2609.31275

[21] Mirko Perkusich, Gustavo Soares, Hyggo Almeida, and Angelo Perkusich. 2015. A procedure to detect problems of processes in software development projects using Bayesian networks. *Expert Systems with Applications* 42, 1 (2015), 437–450. doi:10.1016/j.eswa.2014.08.015

[22] Kumar Rahul and Shovan Chowdhury. 2026. Human–AI Construction of Bayesian Networks for Operational Decision Support: A Virtual Survey Approach. *arXiv preprint arXiv:2607.14141* (2026). doi:10.48550/arXiv.2607.14141

[23] Thiago Rique, Mirko Perkusich, Emanuel Dantas, Danyllo Albuquerque, Kyller Gorgónio, Hyggo Almeida, and Angelo Perkusich. 2023. On Adopting Software Analytics for Managerial Decision-Making: A Practitioner's Perspective. *IEEE Access* 11 (2023), 73145–73163. doi:10.1109/ACCESS.2023.3294823

[24] Thiago Rique, Mirko Perkusich, Kyller Gorgónio, Hyggo Almeida, and Angelo Perkusich. 2025. Constructing the graphical structure of expert-based Bayesian networks in the context of software engineering: A systematic mapping study. *Information and Software Technology* 177 (2025), 107586. doi:10.1016/j.infsof.2024.107586